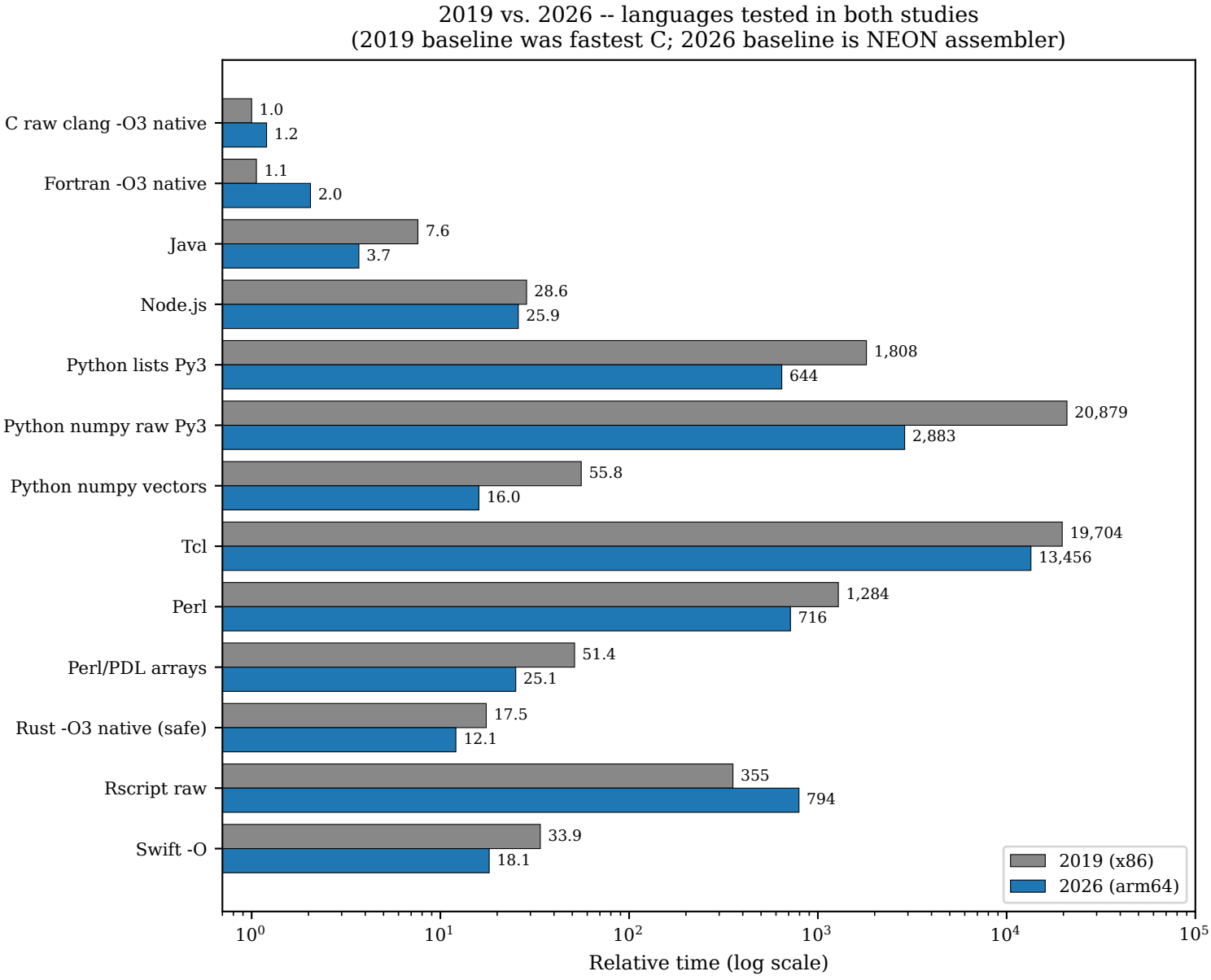
2019 vs. 2026 -- languages tested in both studies
(2019 baseline was fastest C; 2026 baseline is NEON assembler)
C raw clang -O3 native
1.0
1.2
Fortran -O3 native
1.1
2.0
Java
7.6
3.7
Node.js
28.6
25.9
Python lists Py3
1,808
644
Python numpy raw Py3
20,879
2,883
Python numpy vectors
55.8
16.0
Tcl
19,704
13,456
Perl
1,284
716
Perl/PDL arrays
51.4
25.1
Rust -O3 native (safe)
17.5
12.1
Rscript raw
355
794
Swift -O
33.9
18.1
2019 (x86)
2026 (arm64)
10^0
10^1
10^2
10^3
10^4
10^5
Relative time (log scale)